HH-B-671f
January 3, 1966

SUPERSEDING
Int. Fed. Spec. HH-B-00671e (GSA-FSS)
July 1, 1965 and
Fed. Spec. HH-B-671d
August 11, 1955

FEDERAL SPECIFICATION

BRICKS, REFRACTORY: FIRE CLAY

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers fire clay refractory brick.

1.2 Classification.

1.2.1 *Classes.* The brick shall be of the following classes, as specified (see 6.2):

1—Backup duty.
2—Low duty.
3—Slag-resistant duty.
4—High duty.
5—Super duty.

1.2.2 *Dimensions and shapes.* Dimensions and shapes shall be in accordance with figure 1 or special shapes, as specified (see 6.2 and 6.3).

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

Federal Specifications:

QQ-S-781—Steel, Strapping, Flat.
UU-T-111—Tape; Paper, Gummed (Sealing and Securing).
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Box, Fiberboard.
PPP-F-320—Fiberboard; Sheet, Stock and Cut Shapes.
PPP-P-291—Paperboard, Wrapping, Cushioning.
PPP-T-76—Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

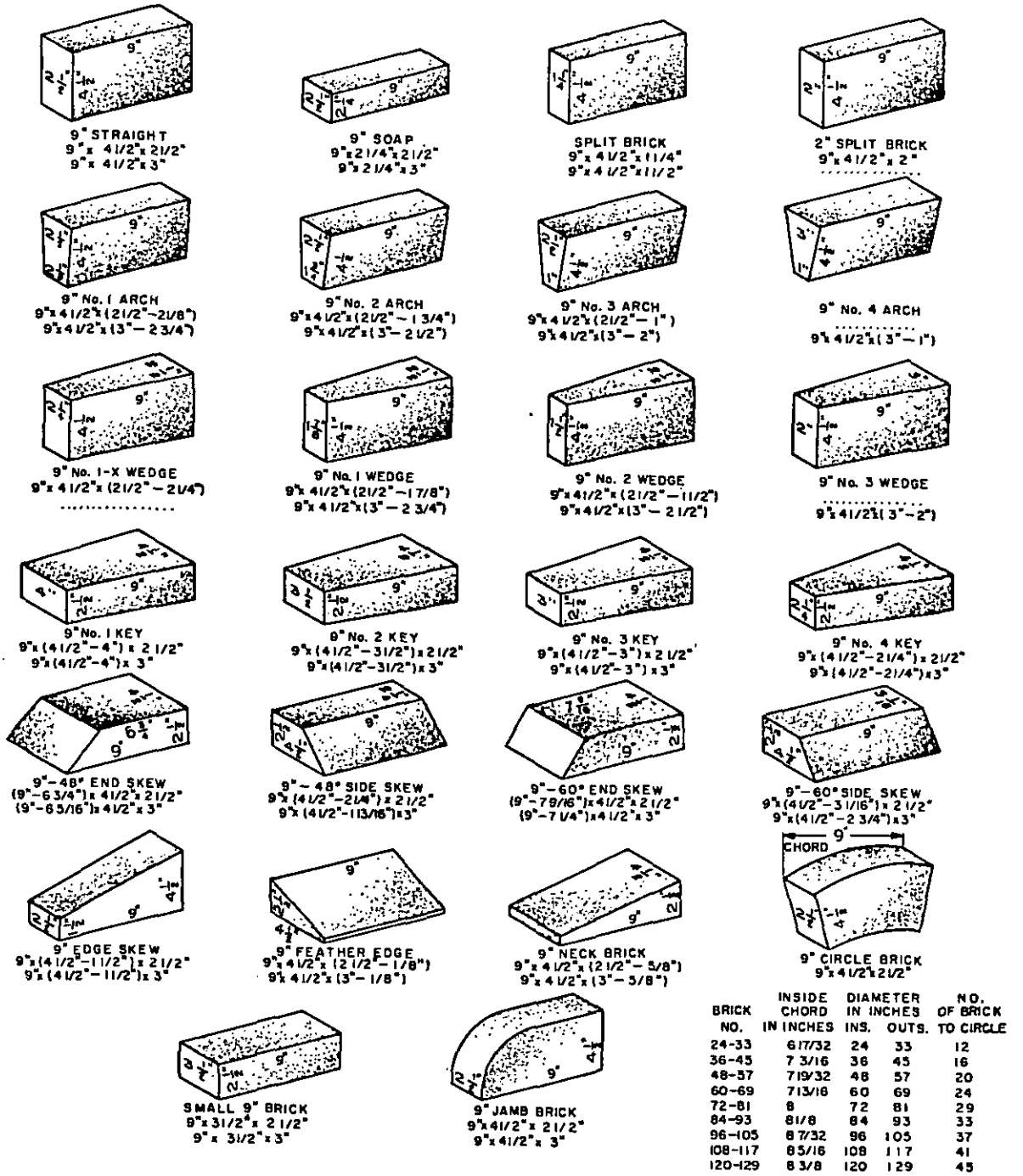
(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

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BRICK NO.	INSIDE CHORD IN INCHES	DIAMETER IN INCHES INS.	OUTS.	NO. OF BRICK TO CIRCLE
24-33	6 17/32	24	33	12
36-45	7 3/16	36	45	16
48-57	7 19/32	48	57	20
60-69	7 13/16	60	69	24
72-81	8	72	81	29
84-93	8 1/8	84	93	33
96-105	8 7/32	96	105	37
108-117	8 5/16	108	117	41
120-129	8 3/8	120	129	45

FIGURE 1. Standard fire clay brick; 9" x 4-1/2" x 2-1/2" Series and 9" x 4-1/2" x 3" Series.

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MIL-STD-147—Palletized Unit Loads (40 In. X 48 In. 4-Way Partial and 4-Way Pallets).

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

C 20-46—Methods of Test for Apparent Porosity, Water Absorption, Apparent Specific Gravity, and Bulk Density of Burned Refractory Brick.

C 24-56—Method of Test for Pyrometric Cone Equivalent (PCE) of Refractory Materials.

C 38-58—Method for Basic Procedure in Panel Spalling Test for Refractory Brick.

C 113-61—Method of Test for Reheat Change of Refractory Brick.

C 122-52—Method of Panel Spalling Test for Super Duty Fireclay Brick.

C 133-55—Methods of Test for Cold Crushing Strength and Modulus of Rupture of Refractory Brick and Shapes.

C 134-41—Methods of Test for Size and Bulk Density of Refractory Brick.

C 154-41—Method of Test for Warpage of Refractory Brick and Tile.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa., 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. Unless otherwise specified (see 6.2), fire clay refractory brick shall be furnished in standard 9-inch series conforming to figure 1. The shape of the brick shall be as specified from figure 1 or a special shape as specified (see 6.2 and 6.3).

3.2 Material. Material shall be selected by the supplier and shall be subject to all provisions of this specification.

3.3 Tolerances in size. All bricks of the standard 9-inch series and all special shapes shall not vary more than plus or minus 2 percent from the specified dimensions on dimensions of 4 inches and over. On dimensions under 4 inches, variations shall be not more than plus or minus 3 percent.

3.4 Warpage. Warpage of brick shall be not more than 1 percent of the diagonal used in making the measurements. The special shapes shall be free from swells, warps, twists, or distortions that will prevent laying brick with a maximum joint of 1/8 inch.

3.5 Physical characteristics. Bricks and special shapes shall have physical characteristics in accordance with table I.

TABLE I. *Physical characteristics*

Class	Pyrometric cone equivalent (PCE) (minimum)		Spalling test, loss (maximum)	Absorption (maximum)	Modulus of rupture	Change of volume
	Cone	*°C.*	*Percent*	*Percent*	*Lb/sq in.*	*Percent*
1	19	1540	—	—	800	—
2	26	1620	—	—	—	—
3	31-1/2	1700	—	8	—	—
4	31-1/2	1700	10.0	—	—	—
5	33	1745	4.0	—	—	+8 or −8

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3.6 Identification marking. The name of the manufacturer or the manufacturer's trade name or trademark shall be molded in each brick or special shape.

3.7 Workmanship. All corners and edges shall be solid and square. The brick shall be compact, unbroken, of homogenous structure, and free from checks, cracks, voids, laminations, or soft centers.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(*a*) Quality conformance inspection (see 4.3).

(*b*) Inspection of preparation for delivery (see 4.5).

4.3 Quality conformance inspection.

4.3.1 *Batch.* A batch shall consist of bricks of one size, shape, and class, produced in a continuous production period with no change in process or formulation of manufacturer's brand of raw ingredients.

4.3.2 *Sampling.*

4.3.2.1 *Examination.* Sampling for examination shall be in accordance with MIL-STD-105, inspection level I.

4.3.2.2 *Tests.* The number of bricks selected for test from each batch shall be in accordance with table II.

4.3.3 *Examination.* Samples selected in accordance with 4.3.2.1 shall be examined in accordance with 4.4.1. AQL shall be 2.5 percent defective.

4.3.4 *Tests.*

4.3.4.1 *Tolerances and warpage.* Samples selected in accordance with table II shall be tested in accordance with 4.4.2.1 and 4.4.2.2. AQL shall be 2.5 percent defective.

4.3.4.2 *Other tests.* Samples selected in accordance with test No. 3, 4, 5, 6, or 7 of table II shall be tested in accordance with 4.4.2.3, 4.4.2.4, 4.4.2.5, 4.4.2.6, or 4.4.2.7, as applicable. Failure of any test shall be cause for rejection of the batch.

4.4 Inspection procedure.

4.4.1 *Examination.* The bricks shall be examined for the following defects:

101. Shapes not as specified.
102. Identification marking not as specified.
103. Workmanship not as specified.

4.4.2 *Tests.*

4.4.2.1 *Tolerances.* The size of the brick shall be determined in accordance with ASTM C 134-41. Dimensional variance of plus or minus 2 percent on dimensions 4 inches and over or dimensional variance of

TABLE II. *Sampling for tests*

Test No.	Test	Number of bricks to be tested				
		Class 1	Class 2	Class 3	Class 4	Class 5
1.	Tolerances	Per MIL-STD-105, inspection level S-3				
2.	Warpage					
3.	PCE	2	2	2	2	2
4.	Spalling	—	—	—	14	14
5.	Absorption	—	—	5	—	—
6.	Modulus of rupture	5	—	—	—	—
7.	Permanent volumetric change	—	—	—	—	3

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plus or minus 3 percent on dimensions under 4 inches, shall constitute failure of this test.

4.4.2.2 *Warpage.* Brick warpage shall be determined in accordance with ASTM C 154-41. Warpage of standard brick exceeding 1 percent of the diagonal used in making measurement, or swells, warps, twists, or distortions of special shapes that will prevent laying of brick with a maximum joint of 1/8 inch shall constitute failure of this test.

4.4.2.3 *Pyrometric cone equivalent.* The PCE shall be determined in accordance with ASTM C 24-56. A cone having a PCE that is less than the PCE specified in table I shall constitute failure of this test.

4.4.2.4 *Spalling resistance.* The spalling resistance of classes 4 and 5 brick shall be determined in accordance with ASTM C 38-58 and ASTM C 122-52. A spalling loss greater than the spalling loss specified in table I shall constitute failure of this test.

4.4.2.5 *Absorption.* The absorption of class 3 brick shall be determined as received, without additional high temperature heat treatment in the laboratory, in accordance with ASTM C 20-46. An absorption of more than 8 percent for class 3 brick shall constitute failure of this test.

4.4.2.6 *Modulus of rupture.* The modulus of rupture of class 1 brick shall be determined in accordance with ASTM C 133-55. A modulus of rupture less than 800 p.s.i. for class 1 brick shall constitute failure of this test.

4.4.2.7 *Permanent volumetric change.* The permanent volumetric change of class 5 brick shall be determined in accordance with ASTM C 113-61, schedule C. The volumes of test specimens shall be determined by calculation from the average of three linear measurements each between the edges, ends, and faces of the brick. The measurements shall be taken so as to be truly representative of the distances between the respective surfaces. A change in volume that is outside the range of plus 8 through minus 3 percent shall constitute failure of this test.

4.5 Inspection of preparation for delivery.

4.5.1 *Quality conformance inspection of pack.*

4.5.1.1 *Unit of product.* For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 *Sampling.* Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 *Examination.* Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective:

104. Materials not as specified for level A or B.
105. Bricks not packaged and packed as specified for level A or B.
106. Marking illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

For civil agency procurement, the definitions and applications of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 *Level A.* Packages shall contain brick of one class, size, and shape only.

5.1.1.1 *Straights.* Straights shall be furnished in packages of 10 bricks. Each package shall contain five courses of two bricks each, stacked in a crisscross formation. Each course of brick shall be separated by a sheet of solid or corrugated fiberboard. Corners of each package shall be protected by scored fiberboard sheet, extending the height of the package and minimum of 1-1/2 inches along each face. The top and bottom of each unit package shall

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be provided with fiberboard caps. Separator sheets, fiberboard caps, and corner protectors shall conform to PPP-F-320, W5c or W5s. Each package shall be tension tied with three straps conforming to QQ-S-781, type I, class B, size 3/8 by 0.020 inch. Two straps shall be placed lengthwise (one on each side of the package) and one strap girthwise (in the center of the package).

5.1.1.1.1 *Alternate method.* Straights shall be packaged in close-fitting boxes conforming to PPP-B-636, V3s or V3c. Each package shall contain five courses of two bricks each, stacked in a crisscross formation. Each course of brick shall be separated by a sheet of solid or corrugated fiberboard. The box shall be closed and reinforced with bands of filament tape in accordance with the appendix to the box specification.

5.1.1.2 *Splits and soaps.* Splits and soaps shall be packaged as specified in 5.1.1.1 or 5.1.1.1.1 except that each package shall consist of 20 bricks arranged in five courses of four bricks each.

5.1.1.3 *Other shape brick and brick not warranting palletization.* Other shape brick and brick procured in quantities not warranting palletization will not require packaging but shall be packed directly into containers specified in 5.2.1.2.

5.1.1.3.1 *Alternate method.* Each brick shall be completely wrapped and all voids shall be filled with paperboard cushioning material conforming to PPP-P-291. Wrapping shall be adequately secured by use of tape conforming to PPP-T-76 or UU-T-111.

5.1.2 *Level B.* Packaging for level B shall be as specified for level A except that the box specified in 5.1.1.1.1 shall be type CF, class domestic, style optional having a minimum 275-pound bursting strength.

5.1.3 *Level C.* The brick shall be packaged to afford protection against damage from the supply source to the first receiving activity in accordance with the recommendation of the supplier.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* Unless otherwise specifed in the contract or order, pallets and shipping containers shall contain brick of one class, size, and shape only.

5.2.1.1 *Straights, splits, and soaps.* Packaged straights, splits, and soaps shall be palletized in accordance with MIL-STD-147.

5.2.1.2 *Other shape brick and brick not warranting palletization.* Other shape brick and brick procured in quantities not warranting palletization shall be packed in a close-fitting box conforming to PPP-B-621, class 2, style 2. The gross weight shall not exceed 500 pounds. The box shall be modified by the addition of skids in accordance with the box specification. Straw, excelsior, or equivalent material shall be used to fill all voids within the box to prevent damage from movement. Strapping shall be zinc coated and placed adjacent to the skids.

5.2.2 *Level B.* Packing for level B shall be as specified for level A except the box specified in 5.2.1.2 shall be class 1 and strapping need not be zinc coated. When the quantity of brick procured is so small that packing of the brick in a wood box would be impractical, the brick shall be packed in a close-fitting box conforming to PPP-B-636, V3c, style optional. The gross weight shall not exceed the weight limitation of the box specification. The box shall be closed and reinforced with bands of pressure sensitive adhesive filament tape in accordance with the appendix to the box specification.

5.2.3 *Level C.* The brick shall be packed to assure carrier acceptance and safe delivery to destination at lowest rates in compliance with carrier rules applicable to the mode of transportation.

5.3 Marking.

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5.3.1 *Military agencies.* Packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 *Civil agencies.* Shipments shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Intended use.

6.1.1 *Class* 1 (*backup duty*). Backup duty brick is intended for structural use, such as in exterior walls of boilers, heat-treating furnaces, and other furnaces where a fire clay brick may be considered desirable but where the brick does not come in direct contact with the heat.

6.1.2 *Class* 2 (*low duty*). Low duty brick is intended for use at moderate temperatures, such as are encountered in boilers operated at average ratings not exceeding 125 percent. Resistance to spalling and slagging is important under these conditions of temperature.

6.1.3 *Class* 3 *(slag-resistant duty)*. Slag-resistant duty brick is intended for use under conditions where resistance to spalling is not of great importance and where resistance to slagging and high temperature is important. In general boiler practice, these bricks may be used in the side walls, but if the refractories used are limited to one brand, material of the high heat duty class is recommended.

6.1.4 *Class* 4 (*high duty*). High duty brick is intended for use under fairly severe conditions of boiler and furnace practice, such as are encountered in plant installations designed to operate at an average rating of over 125 percent and in general incinerator practice. High resistance to slagging, to spalling, and to high temperatures is important.

6.1.5 *Class* 5 (*super duty*). Super duty brick is intended for use under extremely severe service conditions for fire clay brick such as are encountered in oil-fired marine boilers and in plant installations where the use of bricks of unusually high resistance to spalling or high resistance to spalling and volume stability is a major importance, or where volume stability, resistance to high temperature, and severe slagging are also equally important.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Class required (see 1.2.1).
(*c*) Dimensions and shape required (see 3.1).
(*d*) Level of packaging and level of packing required (see 5.1 and 5.2). Level B packaging is intended to provide economical but limited protection and should be specified only when it is determined the bricks will be held in covered storage.
(*e*) Whether pallets and shipping container shall contain brick of more than one size, shape, and class (see 5.2.1 and 5.2.2).

6.3 Special shapes. Special shapes, though not illustrated on figure 1, may be obtained from manufacturers. When special shapes are specified, the procuring activity should provide the manufacturer with a dimensioned illustration.

6.4 Transportation descriptions. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:
Fire brick shapes.
Carload minimum weight 60,000 pounds.

Motor:
Fire brick shapes.
Truckload minimum weight 60,000 pounds, subject to Rule 115, National Motor Freight Classification.

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MILITARY INTEREST:

Custodians:

Army—MO (ERDL)
Navy—YD
Air Force—69

Review activities:

Army—MO
Navy—YD
Air Force—69
Defense Supply Agency—GS

User activities:

Army—None
Navy—SH, MC
Air Force—None

Preparing activity:

Army—MO (ERDL)

Review/user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DOD Standardization Documents.

Civil interest:

GSA, HEW, DC, JUS, PO

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